

1 Kernels

1. Let $\chi = \mathbb{R}^+$

(a) Verify that $\min(x, y) = \int_0^\infty (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) dt$

Choose $x, y \in \mathbb{R}^+$.

Case 1, $x \geq y$

$$\begin{aligned} \int_0^\infty (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) dt &= \int_0^y (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) dt + \\ \int_y^\infty (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) &= \int_0^y (1)(1) dt + 0 = \int_0^y dt = y \end{aligned}$$

Case 2, $x \leq y$

$$\begin{aligned} \int_0^\infty (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) dt &= \int_0^x (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) dt + \\ \int_x^\infty (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) &= \int_0^x (1)(1) dt + 0 = \int_0^x dt = x \end{aligned}$$

Combining both cases gives the desired result.

(b) Show that $K(x, y) = \min(x, y)$ is pos. definite over $\mathbb{R}^+$.

Case 1, $x_i \leq x_j$ and $\alpha \in \mathbb{R}^n$

In the same manner as above

$$\begin{aligned} \sum_{i,j}^n \alpha_i \alpha_j \int_0^\infty (\mathbb{1}_{t < x_i})(\mathbb{1}_{t < x_j}) dt &= \int_0^{x_i} (\sum_i \alpha_i \mathbb{1}_{t < x_i})(\sum_j \alpha_j \mathbb{1}_{t < x_j}) dt = \\ \int_0^{x_i} (\sum_i \alpha_i)(\sum_j \alpha_j) dt &= (\sum_k \alpha_k)^2 x_i \geq 0 \text{ for } x_i \in \mathbb{R}^+. \end{aligned}$$

Likewise for $x_i \geq x_j$,

$$\begin{aligned} \sum_{i,j}^n \alpha_i \alpha_j \int_0^\infty (\mathbb{1}_{t < x_i})(\mathbb{1}_{t < x_j}) dt &= \int_0^{x_j} (\sum_i \alpha_i \mathbb{1}_{t < x_i})(\sum_j \alpha_j \mathbb{1}_{t < x_j}) dt = \\ \int_0^{x_j} (\sum_i \alpha_i)(\sum_j \alpha_j) dt &= (\sum_k \alpha_k)^2 x_j \geq 0 \text{ for } x_k \in \mathbb{R}^+. \end{aligned}$$

Hence K is pos. definite.

(c) Show that $\max(x, y)$ is not pos. definite over $\mathbb{R}^+$.

Consider the case when $n=2$. Let $v = (\alpha_1, \alpha_2)$ and $\gamma = \max(x_1, x_2)$. Then

$$\begin{aligned} v^T K v &= \begin{bmatrix} \alpha_1 & \alpha_2 \end{bmatrix} \begin{bmatrix} x_1 & \gamma \\ \gamma & x_2 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} \\ &= \alpha_1^2 x_1 + 2\alpha_1 \alpha_2 \gamma + \alpha_2^2 x_2 \end{aligned}$$

Since this must hold for any $v \in \mathbb{R}^2$, choose $\alpha_1 = -\alpha_2$.

Then the expression above becomes

$$\begin{aligned} \alpha_1^2 x_1 + 2\alpha_1 \alpha_2 \gamma + \alpha_2^2 x_2 &= \alpha_1^2 (x_1 + x_2 - 2\gamma) = \\ \alpha_1^2 (x_1 + x_2 - 2 \max(x_1, x_2)) &\leq 0 \end{aligned}$$